

Scalable Dual Fingerprinting for Hierarchical Attribution of Text-to-Image Models

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Abstract

The commercialization of generative artificial intelligence (GenAI) has led to a multi-level ecosystem involving model developers, service providers, and consumers. Thus, ensuring traceability is crucial, as service providers may violate intellectual property rights (IPR), and consumers may generate harmful content. However, existing methods are limited to single-level attribution scenarios and cannot simultaneously trace across multiple levels. To this end, we introduce a scalable dual fingerprinting method for text-to-image (T2I) models to achieve traceability of both service providers and consumers. Specifically, we propose 2-head Fingerprint-Informed Low-Rank Adaptation (FI-LoRA), where each head is controlled by a binary fingerprint and capable of introducing the fingerprints into generated images. In practice, one FI-LoRA head is used by the developer to assign a unique fingerprint to each service provider, while the other is made available to service providers for embedding consumer-specific fingerprints during image generation. Experiments show that our method applies to various image generation and editing tasks of multiple T2I models, and can achieve over 99.9% extraction accuracy for both fingerprints. Our method also demonstrates good robustness against both image-level attacks and white-box model-level attacks. Code available at: <https://github.com/jumpycat/FI-LORA>

1. Introduction

With the widespread commercialization of AI models, a layered ecosystem has emerged. The developers (such as research companies) sell or license their models to businesses (service providers). These businesses then integrate the models into applications and subsequently provide con-

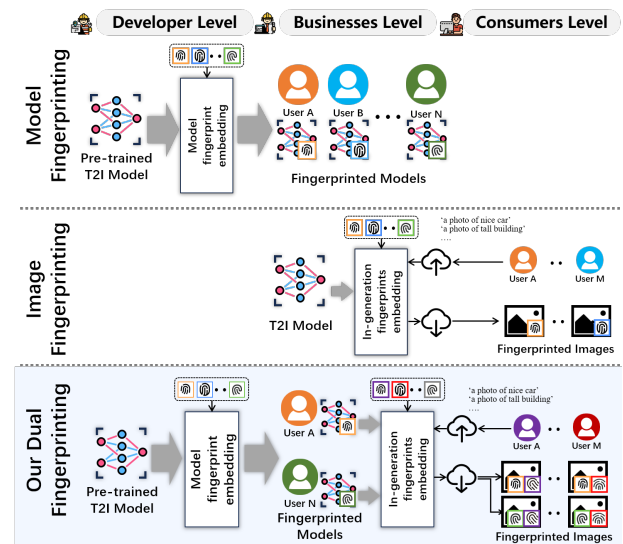


Figure 1. Capability comparison between our dual fingerprinting method and existing model and image fingerprinting methods.

sumers with online services through these applications. For instance, *OpenAI* licenses their GPT models to *Microsoft*, which integrates them into Office products and benefits users with AI-driven features [31]. *Google DeepMind*'s AlphaFold is used by *isomorphic labs* to accelerate drug discovery to help patients [9]. In this ecosystem, the progress in GenAI is particularly outstanding, as diffusion-based image generation and editing models have rapidly become products in various real-world applications, such as Imagen [41], DALL-E 2 [37], GLIDE [34], and Stable Diffusion [35]. For instance, Stability.AI [43] allows businesses to build their applications on top of the Stable Diffusion API and enables artists and designers to generate images. Such an ecosystem in generative AI reduces the technical barriers for businesses and enables them to offer consumers low-cost, user-friendly tools.

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However, as image diffusion models (DM) become widely commercialized and integrated into various applications, concerns regarding their security risks are drawing increasing attention. For example, GenAI businesses may misuse the models that are licensed by the developer for unauthorized activities, such as reselling or operating in violation of licensing terms, while the consumers may be innocent and not responsible. Consumers may use online services provided by the GenAI businesses to create harmful content, such as deepfakes, for which the businesses may not be accountable. Therefore, the need for reliable methods to trace and attribute responsibility or identify illegal roles across the layered ecosystem is critical.

Recent efforts embed imperceptible watermarks into generated images for identification and traceability of their origin. Some research [20, 24, 52, 53] proposed model watermarking methods for image generation models, where the model is optimized so that any generated image includes a watermark. These methods normally serve as solutions for model intellectual property protection, aiming to trace unauthorized redistribution and other model-level infringement activities [13, 15, 29, 32]. When the watermarked models are deployed in online services, these methods cannot trace the responsible consumer since all images share the same fingerprint. Some other research [3, 48, 51] introduced in-generation image watermarking methods by incorporating watermarks during the generation process. This ensures that each generated image contains a unique watermark, which can be used to trace the generation of malicious content. Such methods assume that the businesses and the model owner are the same entity, however, when the model is authorized by another developer, tracing business level infringement becomes unfeasible, as the model does not contain a business-specific watermark.

Thus, current methods focus on different scenarios and have an application limitation. As illustrated in Fig. 1, these methods are limited to tracking only one role, either model providers or consumers, limiting their ability to attribute responsibility across multiple roles. However, in a complex ecosystem where model ownership and content creation roles are distinct, a single watermark fails to simultaneously and hierarchically track both entities.

To bridge this gap, we pioneer a novel hierarchical attribution framework. Unlike existing methods, as shown in Fig. 1, our method introduces dual fingerprinting: business-level fingerprints (managed by the model developer to track authorized deployments) and consumer-level fingerprints (dynamically embedded by business during generation). These fingerprints coexist through a layered embedding strategy, while allowing simultaneous accurate extraction. Our method shares the same objective as the traditional fingerprinting system, and focuses on the embedding of distinct identification fingerprints into different copies, includ-

ing models and images. However, dual fingerprinting is beyond embedding two fingerprints, it requires **(1) different roles should sequentially embed fingerprints that can co-exist**, and **(2) each layer of roles can control one fewer fingerprint than the previous layer**. This implies that, under the 3 role scenario we consider, developers have control over two fingerprints, business manages one, while consumers have no control over any of them.

In particular, we achieve this by the proposed 2-head Fingerprint-Informed Low-Rank Adaptation (FI-LoRA), which enables developers to fine-tune the VAE (Variational AutoEncoder) in the pre-trained T2I models [39], to inject two unique fingerprints into the model weights and allow both to be introduced and coexist in the generated images. One FI-LoRA head is used by developers to trace their businesses, and the other is assigned to the businesses to trace their consumers. Our method enables the use of consumer fingerprint decoders and business fingerprint decoders, respectively, to verify the origin of the content. This can determine whether the content is generated by an unauthorized business or traces malicious consumers, thus realizing effective hierarchical attribution. The contributions of our work are as follows:

- We propose Fingerprint-Informed Low-Rank Adaptation, a novel scalable dual fingerprinting method for text-to-image models that allows two distinct fingerprints to co-exist in the generated images to ensure separate traceability of businesses and consumers independently.
- The proposed FI-LoRA is a pluggable module and allows developers to efficiently and scalably initialize distinct fingerprinted models to a large number of businesses without significant computational costs.
- Experiments demonstrate that images containing dual fingerprints exhibit virtually no visual difference compared to images without fingerprints. Our method is robust against a wide range of fingerprint removal attacks, including post-processing attacks on images (such as cropping, JPEG, etc) and white-box attacks on the model.

2. Related Work

2.1. Image Diffusion Models

DM [10, 22, 39] have shown impressive generative capabilities on images and other modalities [25, 36, 54, 56]. [42] first introduced diffusion probabilistic models to produce data by iteratively applying noise and then reversing the process [22]. LDM [39] performs the diffusion process in the latent space of a pre-trained VAE [11]. Products such as as Stable Diffusion [2], Imagen [41], DALL-E 2 [37] are all based LDM. With the rapid deployment of LDM, the need to ensure the legality of generated content and to establish accountability has become a pressing issue.

2.2. Traceable GenAI

With the advance of Generative AI (GenAI), the demand for forensic and security capabilities has become increasingly important [7, 8, 12]. Among these, traceability stands out as a key requirement to support accountability and trust in GenAI systems. Recent studies on tracing GenAI include passive and proactive methods. The former determines the origin of generated content without modifying the model or the generation process. These methods perform well in closed-set scenarios [6, 30, 46], and many works have also been proposed for open-set scenarios, focusing on mining discriminative features and metrics to identify unseen generative models [1, 21, 44, 47]. Proactive methods embed watermarks (or fingerprints) in the model or generated content, allowing identification and traceability at later stages [20, 38, 52]. Depending on whether the generation process explicitly takes the watermark as input, they can be classified into generative image watermarking [38, 48] and generative model watermarking [15, 20, 24, 27].

Generative Image Watermarking embeds watermarks by adjusting the generation process, including modifying inputs [48, 51] or intermediate features so that generated images carry the watermark. In Tree-ring [48], a circular pattern is embedded in the frequency domain of the initial noise. Gaussian Shading [51] similarly embeds watermarks into the initial noise but preserves a standard Gaussian distribution, thereby ensuring no performance degradation. Recent works [19, 38, 50] suggest incorporating watermarks as an additional input by auxiliary modules, such as adapters [5, 33], to blend the watermark with latent representations and thus produce watermark-embedded images.

Generative Model Watermarking, as a branch of model watermarking [26], aims to encode the watermark in the generated images by optimizing the model weights, rather than explicitly using the watermark as an additional input during image generation. Some methods fine-tune the pre-trained image generator in a supervised manner using a pre-trained watermarking decoder [13, 17, 20, 27]. However, these methods require re-tuning to embed a different watermark. Consequently, some methods attempt to tie the watermark to the model weights by weight modulation [16, 24, 53] or fingerprint-guided normalization layers [14], to enable more efficient fingerprinting.

2.3. Low-Rank Adaptation

Low-Rank Adaptation (LoRA) is an efficient fine-tuning method of large pre-trained models. It aims to reduce the number of parameters updated by decomposing weights into low-rank matrices:

$$W_0 + \Delta W = W_0 + B \cdot A, \quad (1)$$

where, $A \in \mathbb{R}^{r \times k}$, $B \in \mathbb{R}^{d \times r}$, $r \ll d$. LoRA is pluggable so that ΔW can be separated from the model to enable

sharing and reuse by multiple users [49]. Multiple LoRA modules can be combined to improve performance [18].

3. Methodology

3.1. Objectives and Design Motivation

Our dual fingerprinting method is designed to trace both businesses and consumers within a single solution. Thus, the solution focuses on the following objectives:

(1) **Hierarchical Control.** The developer has authority over the fingerprints of businesses $\mathcal{B}^i$, denoted by $\mathbf{m}_b^i$, where $\mathcal{B}$ is the set of businesses. Businesses are restricted to controlling the fingerprints of consumers $\mathcal{C}^i$, denoted by $\mathbf{m}_c^{j|i}$ that uniquely identifies consumer j -th of i -th business.

(2) **Effectiveness.** Both fingerprints in the generated image $\mathbf{x}^{ij}$ should be independently extractable with high accuracy, by business fingerprint decoder $\mathcal{W}_b$ and consumer fingerprint decoder $\mathcal{W}_c$. This can be formulated as $P(\mathcal{F}_b(\mathbf{x}^{ij}), \mathbf{m}_b^i) \geq \tau$ and $P(\mathcal{F}_c(\mathbf{x}^{ij}), \mathbf{m}_c^{j|i}) \geq \tau$, where P computes number of matching bits and τ is a threshold.

(3) **Efficiency.** Developers must be able to efficiently and scalably initialize and allocate models with distinct fingerprints for different businesses.

(4) **Fidelity.** The quality of images generated by models with dual fingerprints should degrade as little as possible, compared to fingerprint-free models. This can be formulated as $Q(\mathbf{x}^{ij}, \mathbf{x}_{\text{clean}}) \leq \epsilon$, where Q is the image distance measurement such as LPIPS, and ϵ is the maximum acceptable degradation.

(5) **Robustness.** The fingerprints should be extractable after unintentional and malicious removal attacks: $\mathbb{E}_{\mathcal{A}}[\min\{P(\mathbf{m}_b^i, \mathcal{W}_b(\mathcal{A}(\mathbf{x}^{ij}))), P(\mathbf{m}_c^{j|i}, \mathcal{W}_c(\mathcal{A}(\mathbf{x}^{ij})))\}] \geq \tau$, where $\mathcal{A}$ is the collection of attacks.

With the above objectives in mind, we consider a LoRA-based design and encode fingerprints into LoRA weights for seamless integration with model weights. This design offers several key advantages: (1) LoRA serves as a modular, plug-and-play component and allows efficient initialization with various fingerprints and scalable adaptation of model parameters to embed different fingerprints. Unlike prior methods that require re-fine-tuning for each new fingerprint [20], the LoRA-based design eliminates this overhead. (2) Adapter-based methods [5, 38] rely on fingerprint inputs during both fine-tuning and inference. In contrast, the LoRA-based design allows standard inference without requiring additional fingerprint inputs after fine-tuning. (3) The LoRA-merged weights are not easy to reverse-engineer, so that users with white-box access to the model cannot easily manipulate the fingerprints.

3.2. Method Overview

Fig. 2 shows the overall pipeline of our method. The developer fine-tunes its T2I model and decoders using our

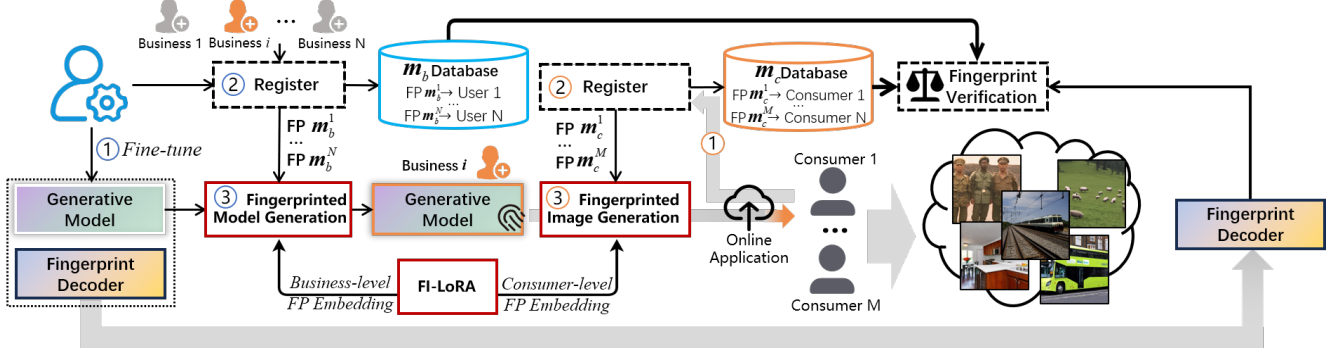


Figure 2. The overall pipeline of our method, where FI-LoRA is the key component to enable hierarchical fingerprinting and attribution.

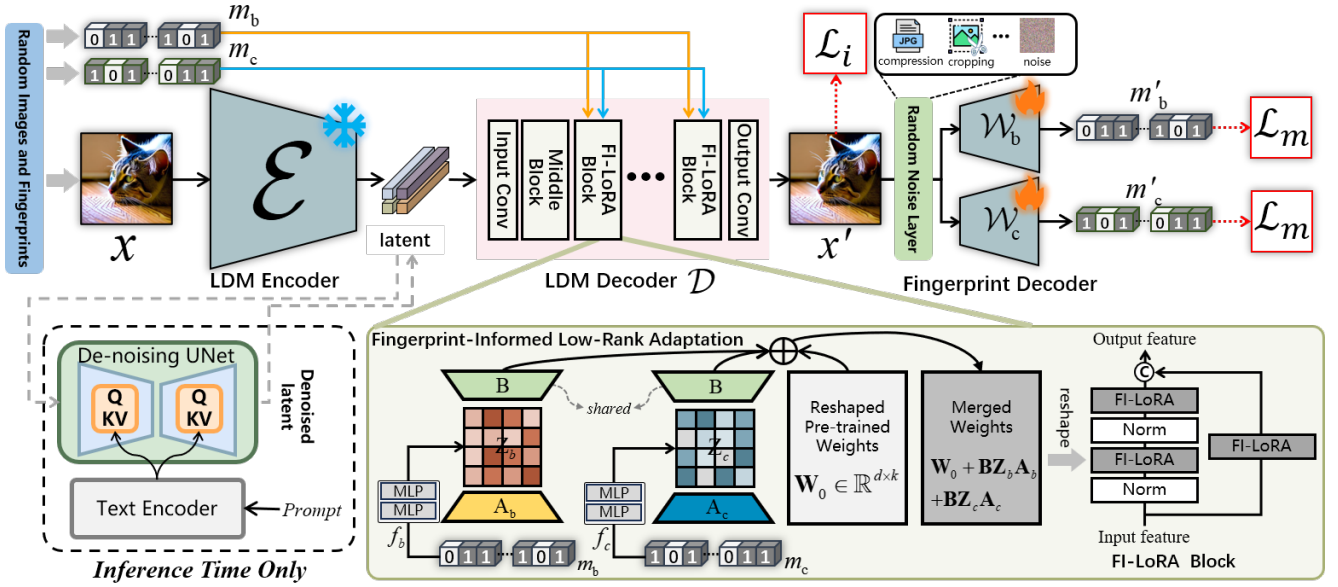


Figure 3. The fine-tuning pipeline of T2I models with FI-LoRA.

method. A unique business-level fingerprint is then assigned to each business entity and registered in a database. This fingerprint is also embedded into the model using FI-LoRA. Within each registered business, consumer-level fingerprints are further incorporated into the image generation process through FI-LoRA. When consumers generate images via the online service of a business, these images will carry unique, identifiable fingerprints that are also registered in a database maintained by the corresponding business. The fingerprint verification enables forensic tracking and accountability, supporting hierarchical attribution from end consumers to businesses.

Below, we provide details of how the developer fine-tunes a pre-trained T2I model using FI-LoRA to enable hierarchical attribution. We take the widely adopted LDM architecture as an example, as illustrated in Fig. 3, we introduce the proposed 2-head FI-LoRA module into the pre-

trained VAE decoder $\mathcal{D}$, and freeze the VAE encoder $\mathcal{E}$. We start by the details of FI-LoRA.

Fingerprint-Informed Low-Rank Adaptation. To encode fingerprints into model weights, we introduce an additional low-rank weight matrix $\mathbf{Z} \in \mathbb{R}^{r \times r}$ between original LoRA weights $\mathbf{A} \in \mathbb{R}^{r \times k}$ and $\mathbf{B} \in \mathbb{R}^{d \times r}$. The weight matrix $\mathbf{Z}$ is derived by a binary fingerprint $\mathbf{m} \in \{0, 1\}^{d_m}$ through an encoding network $f : \{0, 1\}^{d_m} \rightarrow \mathbb{R}^{r \times r}$, and can be formed as $\mathbf{Z} = f(\mathbf{m})$. The final fingerprinted weight $\mathbf{W}_m$ of FI-LoRA is then:

$$\mathbf{W}_m = \mathbf{W}_0 + \Delta \mathbf{W} = \mathbf{W}_0 + \mathbf{B} \mathbf{Z} \mathbf{A}. \quad (2)$$

We observe that $\mathbf{m}$ impacts every element in $\Delta \mathbf{W}$ by interacting with each row and column of matrices $\mathbf{B}$ and $\mathbf{A}$:

$$\Delta \mathbf{W}_{i,j} = \sum_{k=1}^r \sum_{l=1}^r \mathbf{B}_{i,k} \cdot \mathbf{Z}_{k,l} \cdot \mathbf{A}_{l,j}, \quad (3)$$

which helps to fully encode fingerprints into the model weights $\mathbf{W}_m$. $\mathbf{Z}$ can serve as a plug-and-play weight that encodes different fingerprints into the weights $\mathbf{W}_m$.

2-head FI-LoRA. To embed two distinct fingerprints into the weights, we design a 2-head FI-LoRA module that can encode fingerprints $\mathbf{m}_b$ and $\mathbf{m}_c$ into $\mathbf{W}_0$ separately. The 2-head FI-LoRA shares the same $\mathbf{B}$ matrix while has their own $\mathbf{A}_b$, $\mathbf{A}_c$ matrix, and fingerprint encoding network f_b and f_c . Therefore, we have

$$\mathbf{W}_0 + \Delta\mathbf{W} = \mathbf{W}_0 + \mathbf{B}f_b(\mathbf{m}_b)\mathbf{A}_b + \mathbf{B}f_c(\mathbf{m}_c)\mathbf{A}_c. \quad (4)$$

With the proposed 2-head FI-LoRA, developers can seamlessly integrate $\mathbf{B}f_b(\mathbf{m}_b)\mathbf{A}_b$ into the pre-trained weights and reserve $\mathbf{B}f_c(\mathbf{m}_c)\mathbf{A}_c$ for business to engage in consumer fingerprinting. This can prevent businesses from knowing or tampering with the fingerprint $\mathbf{m}_b$.

Fine-tuning Objectives. The 2-head FI-LoRA enables the embedding of fingerprints into the model weights, thus the fine-tuning objective aims to enable the VAE decoder with 2-head FI-LoRA to generate images that contain the fingerprint used in 2-head FI-LoRA, while preserving the image quality. This requires optimizing both image reconstruction and fingerprint decoding accuracy. Below, we introduce the fine-tuning process for the 2-head FI-LoRA.

During the fine-tuning, we randomly sample an image $\mathbf{x}$ and two different fingerprints $\mathbf{m}_b, \mathbf{m}_c \in \{0, 1\}^{d_m}$, as show in Fig. 3. The frozen VAE encoder $\mathcal{E}$ encodes the image in a latent representation $\mathcal{E}(\mathbf{x})$, which will then be fed into the decoder $\mathcal{D}$. Simultaneously, the fingerprints $\mathbf{m}_b$ and $\mathbf{m}_c$ are input into the 2-head FI-LoRA module, generating 2 intermediate codes $\mathbf{Z}_b$ and $\mathbf{Z}_c$, which are combined with 2 sets of LoRA weights to obtain $\mathbf{W}_b = \mathbf{B}\mathbf{Z}_b\mathbf{A}_b$ and $\mathbf{W}_c = \mathbf{B}\mathbf{Z}_c\mathbf{A}_c$, respectively.

$\mathbf{W}_b$ and $\mathbf{W}_c$ are then added to the pre-trained weights $\mathbf{W}_0$, and the merged weights $\mathbf{W}' = \mathbf{W}_0 + \mathbf{W}_b + \mathbf{W}_c$ have the same shape as the original $\mathbf{W}_0$, thus the VAE decoder $\mathcal{D}$ can perform the same convolutional operations without further modification. Finally, the decoder converts the intermediate encoding into an image $\mathbf{x}' = \mathcal{D}(\mathcal{E}(\mathbf{x}))$. Meanwhile, we build two independent fingerprint decoders $\mathcal{W}_b$ and $\mathcal{W}_c$, each taking as input the output image $\mathbf{x}'$ and predict two separate fingerprints $\mathcal{W}_b(\mathbf{x}')$ and $\mathcal{W}_c(\mathbf{x}')$. To enhance the robustness of images against various processing, we follow recent practice [24, 43] and introduce an additional attack simulation layer before the decoder. This layer augments the generated images with augmentations, including JPEG compression, noise, blur, etc.

Fine-tuning involves two optimization objectives: image reconstruction and fingerprint decoding. The loss of image reconstruction is calculated by the difference between the input image $\mathbf{x}$ and the image reconstructed by the decoder using learned perceptual image patch similarity (LPIPS) [57] and mean square error (MSE):

$$\mathcal{L}_i = \mathbb{E}_{\substack{\mathbf{x} \sim \mathcal{X} \\ \mathbf{m}_b, \mathbf{m}_c \sim \{0, 1\}^{d_m}}} \left[\lambda_1 \|\mathbf{x} - \mathcal{D}(\mathcal{E}(\mathbf{x}); \mathbf{m}_b, \mathbf{m}_c)\|_2^2 + \lambda_2 \text{LPIPS}(\mathbf{x}, \mathcal{D}(\mathcal{E}(\mathbf{x}); \mathbf{m}_b, \mathbf{m}_c)) \right], \quad (5)$$

where $\mathcal{D}(\mathcal{E}(\mathbf{x}); \mathbf{m}_b, \mathbf{m}_c)$ denotes the image generated by $\mathcal{D}$ with 2head FI-LoRA that is parameterized with specific $\mathbf{m}_b$ and $\mathbf{m}_c$. In terms of fingerprint decoding, we compute the loss by average bit-wise binary cross-entropy between the ground-truth fingerprints and the predicted fingerprints:

$$\begin{aligned} \mathcal{L}_m = & \mathbb{E}_{\substack{\mathbf{x} \sim \mathcal{X} \\ \mathbf{m}_b, \mathbf{m}_c \sim \{0, 1\}^{d_m}}} \sum_{i=1}^{d_m} \left[\mathbf{m}_b^i \cdot \log \sigma(\mathcal{W}_b(\mathcal{A}(\mathbf{x}'))) \right. \\ & + (1 - \mathbf{m}_b^i) \cdot \log(1 - \sigma(\mathcal{W}_b(\mathcal{A}(\mathbf{x}')))) \\ & + \mathbf{m}_c^i \cdot \log \sigma(\mathcal{W}_c(\mathcal{A}(\mathbf{x}'))) \\ & \left. + (1 - \mathbf{m}_c^i) \cdot \log(1 - \sigma(\mathcal{W}_c(\mathcal{A}(\mathbf{x}')))) \right], \end{aligned} \quad (6)$$

where $\mathcal{A}$ is the attack simulation layer. Then, the overall fine-tuning objective is:

$$\mathcal{L} = \mathcal{L}_i + \lambda_3 \mathcal{L}_m, \quad (7)$$

where λ_1 , λ_2 , and λ_3 are hyperparameters that balance the different loss terms, and we set the same λ_3 for $\mathbf{m}_b$ and $\mathbf{m}_c$. As shown in Fig. 3, the U-Net is not involved in the fine-tuning and is exclusively used in inference time. The de-noised latent image representation is forwarded to the fine-tuned VAE decoder, which subsequently generates an image containing two fingerprints.

4. Experiments

4.1. Dataset, Models, Tasks, and Settings

Datasets. We used the training set of MS-COCO-2017 [28] to fine-tune the LDM decoder $\mathcal{D}$. The evaluations were performed on the MS-COCO-2017 val set, ImageNet [40], MagicBrush [55], and InstructPix2Pix [4]. All images are randomly cropped to 512×512 pixels.

Models. We evaluated our method on Stable Diffusion SD 2 base (SD2-base) and InstructPix2Pix [4]. The fingerprint decoder $\mathcal{W}$ is based on Efficient-b0 [45]. We used 32 bits for both $\mathbf{m}_b$ and $\mathbf{m}_c$, making the total payload 64 bits. Both f_b and f_c employ a 2-layer fully connected network.

Tasks. We studied both T2I and Image-to-Image (I2I) tasks. For T2I, we used captions from MS-COCO-2017 val set and ImageNet as prompts to generate images. For I2I, we used MagicBrush and the InstructPix2Pix [4] dataset for text-guided image editing. For comparison, as our method is classified as model fingerprinting, compare it with methods of the same category, including **Per. Norm** [14], **Sta. Sig** [20], and **WOUAF** [24]. **Per. Norm** [14] introduced personalized normalization for GAN fingerprinting, **Sta. Sig** [20] proposed to watermark LDM with a pre-trained watermarking decoder, **WOUAF** [24] proposed to fingerprint LDM with weight modulation [53].

Table 1. Performances of image reconstruction and generation on four datasets and tasks. (PSNR $\uparrow$; SSIM $\uparrow$; LPIPS $\downarrow$; FID $\downarrow$; Clip $\uparrow$).

Dataset	Method	Image Quality -recon						Image Quality -gen	
		PSNR _b	PSNR _r	SSIM _b	SSIM _r	LPIPS _b	LPIPS _r	FID _r	Clip Score
CoCo (T2I) SD2-base	No fingerprint	-	28.556	-	0.805	-	0.128	23.265	0.658
	<i>Per. Norm.</i> [14]	30.219	28.360	0.864	0.801	<u>0.067</u>	<u>0.124</u>	<u>23.726</u>	0.658
	<i>Sta. Sig.</i> [20]	31.259	28.515	0.899	0.820	0.082	0.142	23.895	0.658
	WOUAF [24]	<u>31.345</u>	28.590	0.887	<u>0.817</u>	0.080	0.129	23.741	0.658
	Ours	32.359	<u>28.541</u>	<u>0.889</u>	0.812	0.066	0.124	23.272	0.658
ImageNet (T2I) SD2-base	No fingerprint	-	29.581	-	0.825	-	0.125	24.239	0.659
	<i>Per. Norm.</i> [14]	30.967	28.540	0.884	0.815	<u>0.064</u>	<u>0.125</u>	<u>25.014</u>	0.659
	<i>Sta. Sig.</i> [20]	32.462	29.112	0.924	0.835	0.080	0.147	25.990	0.659
	WOUAF [24]	<u>32.493</u>	<u>29.210</u>	<u>0.923</u>	0.834	0.081	0.147	25.875	0.659
	Ours	33.343	29.684	0.907	0.831	0.063	0.124	24.240	0.659
MagicBrush (I2I) SD2-inpainting	No fingerprint	-	30.823	-	0.871	-	0.105	19.922	0.620
	<i>Per. Norm.</i> [14]	31.590	29.387	0.903	0.851	<u>0.050</u>	<u>0.105</u>	<u>21.428</u>	0.620
	<i>Sta. Sig.</i> [20]	32.996	30.157	0.931	0.857	0.065	0.118	22.304	0.620
	WOUAF [24]	<u>33.140</u>	<u>30.295</u>	<u>0.932</u>	<u>0.860</u>	0.065	0.115	22.289	0.620
	Ours	35.020	31.014	0.934	0.877	0.048	0.103	19.979	0.620
InstructPix2Pix (I2I) InstructPix2Pix [4]	No fingerprint	-	33.492	-	0.936	-	0.040	13.216	0.657
	<i>Per. Norm.</i> [14]	30.026	29.450	0.858	<u>0.835</u>	0.060	0.071	13.319	0.658
	<i>Sta. Sig.</i> [20]	30.015	29.373	0.829	0.820	0.060	0.071	<u>13.310</u>	0.658
	WOUAF [24]	<u>30.219</u>	<u>29.456</u>	0.831	0.820	0.064	0.072	13.243	0.657
	Ours	30.637	29.813	<u>0.857</u>	0.839	<u>0.061</u>	0.071	12.953	0.658

Evaluation Metrics. We used (1) Peak Signal-to-Noise Ratio (PSNR), Structural Similarity (SSIM), LPIPS, and Fréchet Inception Distance (FID) to measure the quality of images reconstructed and generated by the fingerprinted model. We also used the CLIP Score to quantify the semantic alignment between the prompts and generated images. All metrics are computed based on 3,200 image pairs, except FID, using 10k. As for the fingerprints, we used bit-wise accuracy (Bit Acc) to quantify the accuracy of bit-level matching between the extracted fingerprint $\mathcal{W}(\mathbf{x}')$ and ground truth fingerprint $\mathbf{m}$:

$$\text{Bit Acc} = \frac{1}{d_m} \sum_{i=1}^{d_m} \mathbb{1}(\mathcal{W}(\mathbf{x}')_i = \mathbf{m}_i), \quad (8)$$

where $\mathbb{1}$ is the indicator function that returns 1 when the enclosed condition is satisfied and 0 otherwise. We also calculated the false positive rate (FPR) and true positive rate (TPR) by specific matching thresholds τ . An image is identified as fingerprinted if $\sum_{i=1}^{d_m} \mathbb{1}(\mathcal{W}(\mathbf{x}')_i = \mathbf{m}_i) \geq \tau$; otherwise, it is rejected as a non-fingerprinted.

4.2. Image Quality Analysis

Our method has minimal impact on image quality. We evaluated the image generation performances of our method across four datasets with different models. As shown in Table 1, the evaluation metrics are divided into image recon-



Figure 4. Images generated by the no fingerprint model, by Sta.Sig. / FI-LoRA fingerprinted model using the same prompts. Each column corresponds to a random pair of $\mathbf{m}_b$ and $\mathbf{m}_c$.

struction quality (“_recon”) and image generation quality (“_gen”). The subscript “b” indicates that the metric compares images generated by the baseline (original no fingerprint model) model and fingerprinted model, while “r” denotes comparisons between real images and generated ones.

In terms of image reconstruction, FI-LoRA outperforms the compared methods across all datasets for PSNR_b and achieves an average improvement of 1.041 over WOUAF [24]. For PSNR_r, the improvement averages 0.375. In terms of SSIM, FI-LoRA shows competitive results with an average improvement of 0.001 for SSIM_b and 0.003 for SSIM_r. Similarly, FI-LoRA surpasses the second-best methods by 0.001 in both LPIPS_b and LPIPS_r. In terms of image generation, FI-LoRA achieves the best FID_r

scores across all datasets. It shows improvements over the second-best methods, reducing FID_r by 0.454 on CoCo, 0.774 on ImageNet, 1.449 on MagicBrush, and 0.290 on InstructPix2Pix, even surpassing the baseline by 0.263. These results demonstrate FI-LoRA’s superior strong fidelity in generating highly realistic images across diverse datasets.

In summary, FI-LoRA achieved superior performance in image reconstruction, with even greater advantages in image generation. This is largely due to the higher weight assigned to the LPIPS loss during FI-LoRA fine-tuning, compared to the original VAE training (1:0.1). This results in more improvements in FID, as opposed to MSE-based metrics like PSNR. In Fig. 4, we show images generated using the same prompt and seed by both the original SD2-base and our fingerprinted model. Notably, FI-LoRA effectively preserves the visual appearance and intricate details.

Table 2. Fingerprint performances in terms of Bit Acc, FPR($\tau=28$), and Bit Acc when images are cropped (50%) and JPEG compressed (qf=50).

Datasets	Methods	Acc	FPR	Crop	JPEG
CoCo	<i>Per. Norm.</i> [14]	99.67	0.0	84.43	93.21
	<i>Sta. Sig.</i> [20]	99.52	0.0	83.28	80.18
	<i>WOUAF</i> [24]	99.30	0.0	79.46	97.55
	Ours - m_b	99.96	0.0	86.32	99.29
	Ours - m_c	99.91	0.0	80.88	99.71
ImageNet	<i>Per. Norm.</i> [14]	99.82	0.0	85.16	92.97
	<i>Sta. Sig.</i> [20]	99.52	0.0	84.15	81.46
	<i>WOUAF</i> [24]	99.28	0.0	78.60	97.42
	Ours - m_b	99.93	0.0	85.94	99.32
	Ours - m_c	99.91	0.0	81.24	99.11
Magic Brush	<i>Per. Norm.</i> [14]	99.71	0.0	85.25	93.76
	<i>Sta. Sig.</i> [20]	99.64	0.0	83.55	80.97
	<i>WOUAF</i> [24]	99.32	0.0	78.99	97.45
	Ours - m_b	99.94	0.0	85.80	99.26
	Ours - m_c	99.95	0.0	81.22	99.59
Instruct Pix2Pix	<i>Per. Norm.</i> [14]	99.65	0.0	85.29	93.50
	<i>Sta. Sig.</i> [20]	99.59	0.0	84.23	81.39
	<i>WOUAF</i> [24]	99.35	0.0	76.53	96.82
	Ours - m_b	99.90	0.0	87.14	99.10
	Ours - m_c	99.94	0.0	82.65	99.12

4.3. Fingerprint Effectiveness Analysis

Our method achieves near-perfect accuracy in fingerprint extraction. In Table 2, we present the fingerprinting effectiveness. It is worth highlighting that our images contain two fingerprints, doubling the payload compared to other methods (32×2 vs. 32). We can see that all methods show high performance, with Bit ACC above 99% across datasets. However, in our method, both fingerprints m_b and m_c achieve even higher results. For example, on CoCo,

our m_b reaches 99.96%, surpassing *Per. Norm.* (99.67%) and *WOUAF* (99.30%). This advantage is consistent across other tasks, demonstrating the high effectiveness of our method in fingerprinting. We evaluated the FPR on 32,000 images with a threshold τ of 28. While all methods report a 0.0% FPR, the higher Bit Acc of FI-LoRA indicates a better overall performance. Experiments show that $\tau = 28$ can achieve the best balance between TPR and FPR.

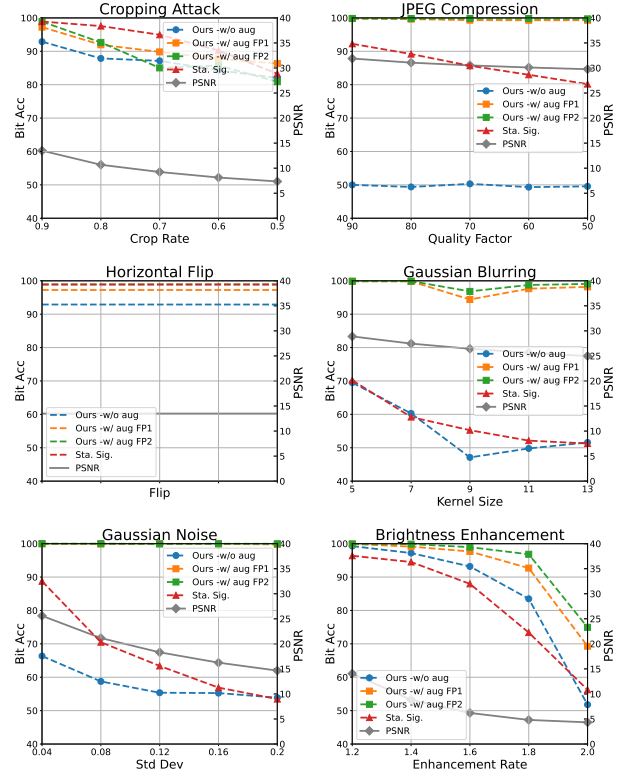


Figure 5. Bit Acc under image-level attacks.

4.4. Fingerprint Robustness Analysis

We evaluated the robustness of FI-LoRA against fingerprint removal attacks. Businesses with white-box model access can launch model-level attacks, while consumers can launch image-level attacks. Robustness is quantitatively measured through Bit Acc under attacks.

Image Processing Attacks aim to remove the fingerprints through post-processing. We evaluated six common types of attacks (e.g., JPEG compression, cropping), as shown in Fig. 5. Each attack was tested at six intensity levels, and we report the PSNR between pre- and post-attack images, along with the Bit Acc of m_b (FP1) and m_c (FP2). We also compared our method with and without attack simulation augmentation (-w/ aug vs. -w/o aug). As attack intensity increases, Bit Acc decreases. However, introducing the noise layer significantly improves the robustness. For instance, under JPEG compression or high-intensity Gaussian noise (Std=0.2), the noise layer improves Bit Acc from 50% to

nearly 100%. Compared to *Sta. Sig.*[20], the attack simulation offers better utility since our jointly optimized decoder effectively learns to extract fingerprints from distorted images. Notably, high-intensity attacks (e.g., 2x brightness) can reduce Bit Acc below 80%, but this causes severe image degradation ($\text{PSNR} \leq 10$), suggesting that removing fingerprints comes at the cost of compromising image quality.

Model Modification Attacks We evaluated the robustness against white-box model-level attacks, including weight pruning, quantization, and noise addition. Pruning zeros out weights with the smallest absolute values, quantization reduces precision by limiting decimal places, and noise addition introduces Gaussian noise to model weights. As shown in Fig. 6, we report the Bit Acc along with PSNR between images generated by the attacked and original models. Our method consistently achieves superior robustness compared to *WOUAF*[24] and *Sta. Sig.*[20]. For example, with Gaussian noise ($\text{std}=0.02$), our Bit Acc remains near 100%, while theirs drops below 70%. Under stronger attacks, such as 50% pruning or noise with std 0.05, our performance is slightly lower than *Per. Norm.*[14] by about 5%, however, at this point, the model functionality decreases significantly, with PSNR below 15.

Recent studies show that generative model watermarking is vulnerable to adversarial fine-tuning attacks [23]. Thus, we evaluated robustness under such attacks, where the model is further fine-tuned by only image loss. As shown in Fig. 6, we observe that as the number of fine-tuning steps increases, the Bit Acc of all compared methods gradually declines, dropping below 70% after 10k steps. While our method also experiences some decrease, the decline stabilizes with more steps, and at 10k steps, it remains above 80%, surpassing the compared methods by over 10%.

We also evaluated collusion attacks among businesses, which is a significant challenge for model watermarking [20]. While our method does not directly achieve robustness, we found that the bias in fingerprint logits of the pirated model provides evidence for traceability.

Limitations. A limitation of FI-LoRA is the capacity: since the generated images contain two fingerprints, the effective bits for each fingerprint are half of the total capacity. Notably, when a single fingerprint has 64 bits, the fingerprinting loss may not converge. However, using two 32-bit fingerprints with hierarchical attribution can still result in promising reliability.

5. Conclusion

We propose a novel scalable dual fingerprinting method for T2I models, designed to achieve traceability across multiple roles in the Gen AI ecosystem. Our proposed 2 head FI-LoRA can successfully embed two independent fingerprints into generated images with minimal impact on image quality and achieve over 99.9% extraction accuracy. We demonstrate that our method enables hierarchical and independent accountability, allowing developers to trace both model leakage at the business level and illegal content generation at the consumer level, with improved robustness against many attacks. Overall, our work provides a signif-

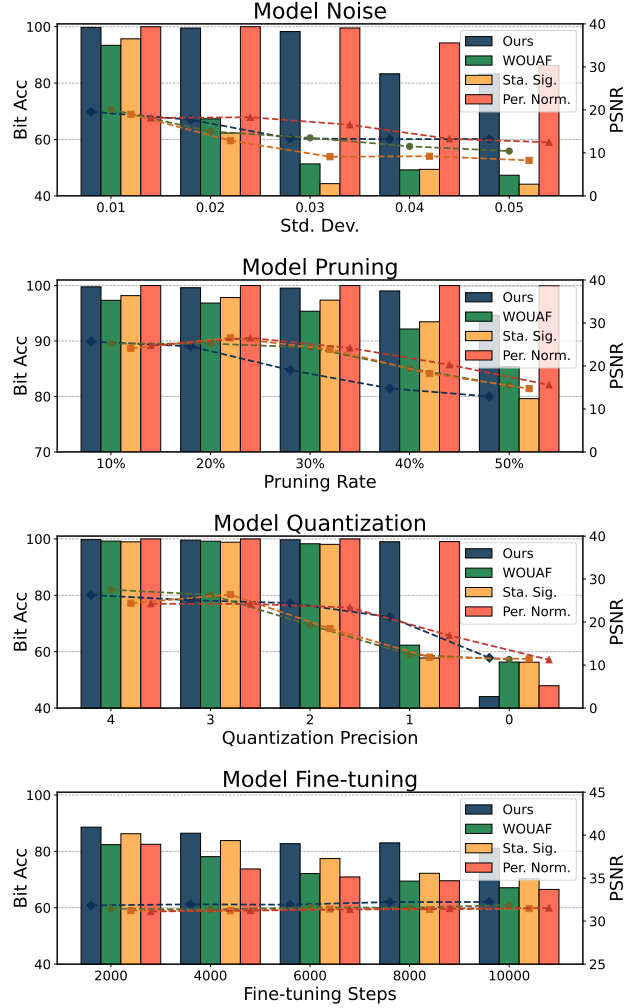


Figure 6. Bit Acc under model-level attacks.

icant step towards ensuring accountability and traceability in the rapidly growing ecosystem of generative AI.

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